

## PART : CHEMISTRY

1. Difference of B.P and M.P in oxygen and sulphur can be explained by  
 (1) Atomicity (2) Atomic mass (3) Electronegativity (4) Electron gain enthalpy

**Ans. (1)**

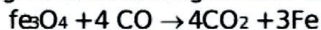
**Sol.** The large difference between the melting and boiling points of oxygen and sulphur may be explained on the basis of their atomicity.  $O_2$  exist as diatomic molecule where sulphur exist as polyatomic molecule  $S_8$

2. Which of the following strong oxidising agent?  
 (1)  $Eu^{+2}$  (2)  $Ce^{2+}$  (3)  $Ce^{4+}$  (4)  $Eu^{4+}$

**Ans. (4)**

**Sol.**  $M^{4+}$  will reduce itself to stable (+3) so, it will be good Oxidizing agent.

3. If 280 kg CO and 2320 kg of  $Fe_3O_4$  are made to react according to



what is the weight of Fe produce (in kg)

Given : Mol. Wt. of CO and  $Fe_3O_4$  are 28 and 232 u

**Ans. (420)**

**Sol.**

|           |   |       |   |                                        |
|-----------|---|-------|---|----------------------------------------|
| $Fe_3O_4$ | + | 4 CO  | → | 3 Fe + 4 CO <sub>2</sub>               |
| n         |   | 10000 |   | 10000 (LR)                             |
|           |   |       |   | $10000 \times \frac{3}{4} = 7500$      |
|           |   |       |   | ∴ $W_{Fe} = 7500 \times 56 g = 420 kg$ |

4. A reaction is non spontaneous at freezing point and spontaneous at boiling point select the correct option

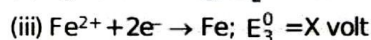
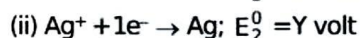
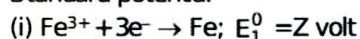
- (1) Both  $\Delta H$  and  $\Delta S$  are positive (2)  $\Delta H > 0$ ,  $\Delta S < 0$   
 (3)  $\Delta H < 0$ ,  $\Delta S > 0$  (4) Both  $\Delta H$  and  $\Delta S$  are negative

**Ans. (1)**

**Sol.**

| Case I                     | Case II                    |
|----------------------------|----------------------------|
| At freezing point          | At boiling point           |
| $\Delta G > 0$             | $\Delta G < 0$             |
| $\Delta G - T\Delta S > 0$ | $\Delta H - T\Delta S < 0$ |
| $ \Delta G  >  T\Delta S $ | $ \Delta H  <  T\Delta S $ |

5. Standard potential



Find  $E^0$  of reaction  $Fe^{2+} + Ag^+ \rightarrow Fe^{3+} + Ag$  (s)

- (1)  $(Z - 2X + 3Y)$  volt (2)  $(X - 2Y + 3Z)$  volt (3)  $(Y - 2Z + 3X)$  volt (4)  $(Y - 3Z + 2X)$  volt

**Ans. (4)**

**Sol.**

$$Fe^{3+} + e^- \rightarrow Fe^{2+}$$

(iv) = (i) - (iii)

$$\Delta G_4^0 = \Delta G_1^0 - \Delta G_3^0$$

$$-1.f.E_4^0 = -3.f.E_1^0 + 2.f.E_3^0$$

$$E_4^0 = 3E_1^0 - 2E_3^0 = 3Z - 2X$$

$$E^0 = E_{Ag}^0 - E_{Fe^{2+}/Fe^{3+}}^0 = Y - E_{Fe^{2+}/Fe^{3+}}^0 = Y - 3Z + 2X$$

6. In a process pressure of gas is directly proportional to temperature then choose correct option  
 (A) Process is isochoric  
 (B) Work done in process is zero  
 (C) Internal energy increases with increase in temperature  
 (1) A and B are correct (2) A and C are correct  
 (3) A, B and C are correct (4) B and C are correct

**Ans. (3)**

**Sol.**  $P \propto T$  ( $V, n = \text{constant}$ )

7. 1 mole of a complex with molecular formula  $\text{Co}(\text{NH}_3)_5\text{Cl}_3$  produces 3 mole ions upon complete ionisation. Upon adding excess  $\text{AgNO}_3$ , 2 mole  $\text{AgCl}$  are precipitated. Complex is:  
 (1)  $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$  (2)  $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl} \cdot \text{NH}_3$   
 (3)  $[\text{Co}(\text{NH}_3)_3\text{Cl}_3] \cdot 2\text{NH}_3$  (4)  $[\text{Co}(\text{NH}_3)_4\text{Cl}]\text{Cl}_2 \cdot \text{NH}_3$

**Ans. (1)**

**Sol.**  $2\text{Cl}^-$  should be outside square bracket. Also,  $\text{Co}^{3+}$  shows  $\text{CN} = 6$   
 $\therefore$  Complex :  $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$

8. Consider  $\text{H}_2\text{O}$ ,  $\text{NH}_3$ ,  $\text{CH}_4$  and select correct statements:

- (A) All are  $\text{sp}^3$  hybridised  
 (B) Bond angle  $\text{H}-\text{O}-\text{H}$ ,  $\text{H}-\text{N}-\text{H}$ ,  $\text{H}-\text{C}-\text{H}$  are  $104.5^\circ$ ,  $107^\circ$ ,  $109.5^\circ$   
 (C) Dipole moment:  $\text{CH}_4 < \text{NH}_3 < \text{H}_2\text{O}$   
 (D)  $\text{H}_2\text{O}$  and  $\text{NH}_3$  are lewis base and lewis acid  
 (E)  $\text{NH}_3$  in  $\text{H}_2\text{O}$  is basic in nature  
 (1) A, B, C and E (2) A, B and D (3) A, C, D (4) B, D, E

**Ans. (1)**

**Sol.** Both  $\text{NH}_3$  and  $\text{H}_2\text{O}$  behave as lewis base.

9. Which of the following is not true combination from given statement.  
 (a) Elements in periodic table are linearly arranged with atomic weight  
 (b) Elements in periodic table are linearly arranged with atomic number  
 (c) Element having similar electronic configuration are arranged in same group  
 (d) Using periodic table we can identify in which subshell last electron enters  
 (e) Isotopes of an element are placed in periodic table.  
 (1) a, e only (2) b, e only (3) b, c, d only (4) a, c, d, e only

**Ans. (1)**

10. If the  $K_{\text{sp}}$  of  $\text{Cr}(\text{OH})_3$  is  $1.6 \times 10^{-30} \text{ M}^4$  the molar solubility of salt in water is  
 (1)  $\left(\frac{1.6 \times 10^{-30}}{27}\right)^{1/4}$  (2)  $\left(\frac{16}{27} \times 10^{-30}\right)^{1/4}$  (3)  $\left(\frac{160}{27} \times 10^{-30}\right)^{1/4}$  (4) None of these

**Ans. (1)**

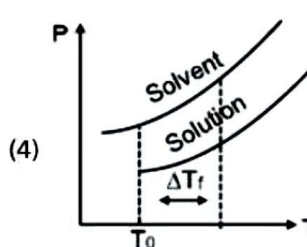
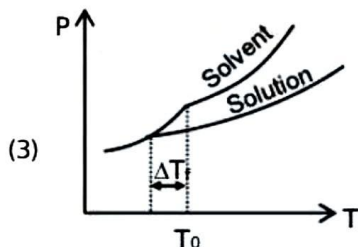
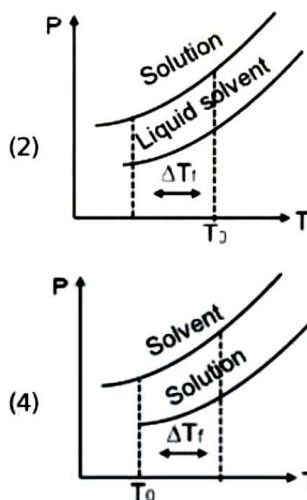
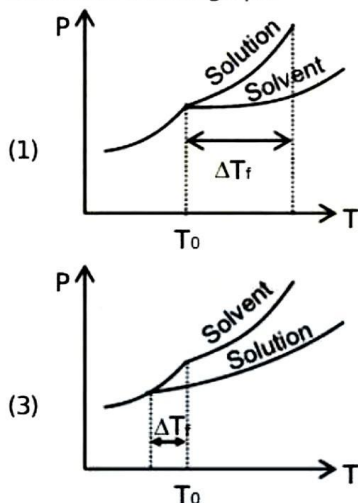
**Sol.**  $\text{Cr}(\text{OH})_3(\text{s}) \rightleftharpoons \underset{\text{s}}{\text{Cr}^{3+}} + \underset{3\text{s}}{3\text{OH}^-}$

$$K_{\text{sp}} = [\text{Cr}^{3+}][\text{OH}^-]^3 = s(3s)^3$$

$$= 27s^4$$

$$s = \left(\frac{1.6 \times 10^{-30}}{27}\right)^{1/4}$$

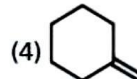
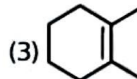
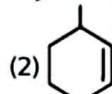
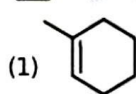
11. Select the correct graph.



Ans. (3)

Sol. T at which  $VP_{\text{solid}} = VP_{\text{liquid}}$  is freezing point of solution is less than that of solvent.

12.  is formed by ozonolysis followed by aldol condensation of Alkene. Alkene can be:



Ans. (1)

13. Ribose present in DNA  
 (A) It is a pentose sugar  
 (B) It is a present in pyranose form  
 (C) It is present in D-configuration  
 (D) It is reducing sugar in free form  
 (E)  $\alpha$  anomeric form is present.

Correct options are :

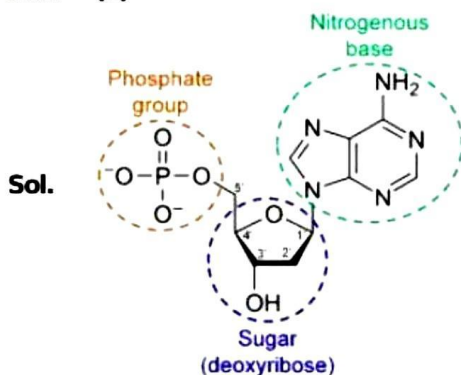
(1) A, C, D

(2) A, B, D

(3) A, B, C, D, E

(4) A, C, E

Ans. (1)



14. Arrange following for reaction rate with nucleophilic attack  
 (a) Acetophenone (b) p-tolylaldehyde  
 (c) Benzaldehyde (d) p-Nitrobenzaldehyde  
 (1)  $d > c > b > a$  (2)  $a > c > b > d$  (3)  $d > b > c > a$  (4)  $d > a > b > c$

Ans. (1)

15. Which compound react fastest with HBr?



Ans. (1)

16. Stability of carbocation is maximum is?



Ans. (2)

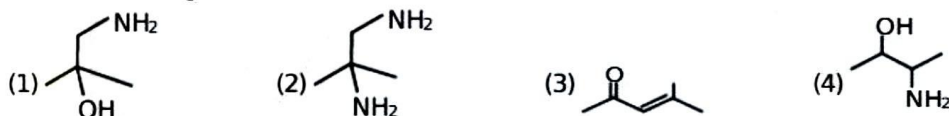
17. **Statement-I** : Dumas method is used for estimation of Nitrogen.

**Statement-II** : In Dumas method Nitrogen present in compound is converted to  $(\text{NH}_4)_2\text{SO}_4$

- (1) Both Statement I and statement II are true  
 (2) Both statement I and statement II are false  
 (3) Statement I is true but statement II is false  
 (4) Statement I is false but statement II is true

Ans. (3)

18.  $\text{CH}_3\text{-C}\equiv\text{CH} \xrightarrow[3. \text{H}_2/\text{Ni}]{1. \text{HgSO}_4, 2. \text{HCN}/\text{HO}^\ominus}$



Ans. (1)

19. **Statement-I** :  $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CH}_2\text{-Cl} + \text{OH}^\ominus \longrightarrow$  Reaction is favoured in less polar solvent.

**Statement-II** :  $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CH}_2\text{-Cl} + \text{R}_3\text{N} \longrightarrow$  Reaction is favoured in more polar solvent.

- (1) Both Statement I and statement II are true (2) Both statement I and statement II are false  
 (3) Statement I is true but statement II is false (4) Statement I is false but statement II is true

Ans. (1)